

By Newton's 3rd Law, force exerted by car has same magnitude as the normal contact force

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$$g = \frac{GM}{R^2}$$

$$\begin{aligned}\text{Work done} = \text{increase in GPE} &= -\frac{GMm}{3R} - \frac{GMm}{R} \\ &= \frac{2}{3} \frac{GMm}{R} \\ &= \frac{2}{3} \frac{GMm}{R^2} R \\ &= \frac{2}{3} mgR\end{aligned}$$

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